

4.13.16

EE25BTECH11011 - Navya Banavathu

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Question

The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for

- ① exactly one value of p
- ② exactly two values of p
- ③ more than two values of p
- ④ no value of p

Solution

Given lines are,

$$p(p^2 + 1)x - y + q = 0 \quad (1)$$

$$(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0 \quad (2)$$

The normal vectors of the lines are

$$\mathbf{n}_1 = \begin{pmatrix} p(p^2 + 1) \\ -1 \end{pmatrix}, \quad \mathbf{n}_2 = \begin{pmatrix} (p^2 + 1)^2 \\ p^2 + 1 \end{pmatrix} \quad (3)$$

If both lines are perpendicular to a common line with normal vector $\mathbf{m}$, then

$$\mathbf{n}_1^T \mathbf{m} = 0, \quad (4)$$

$$\mathbf{n}_2^T \mathbf{m} = 0. \quad (5)$$

Solution

Hence, the two equations can be written together as

$$\begin{pmatrix} p(p^2 + 1) & -1 \\ (p^2 + 1)^2 & p^2 + 1 \end{pmatrix} \mathbf{m} = \mathbf{0}. \quad (6)$$

Writing the system as an augmented matrix:

$$\left[\begin{array}{cc|c} p(p^2 + 1) & -1 & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \quad (7)$$

Solution

Perform row operations to reduce to RREF:

$$\left[\begin{array}{cc|c} p(p^2 + 1) & -1 & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow \frac{1}{p(p^2+1)} R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ (p^2 + 1)^2 & p^2 + 1 & 0 \end{array} \right] \quad (8)$$

$$\xrightarrow{R_2 \rightarrow R_2 - (p^2 + 1)^2 R_1} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ 0 & (p^2 + 1) + \frac{(p^2+1)^2}{p(p^2+1)} & 0 \end{array} \right] \quad (9)$$

$$= \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ 0 & (p^2 + 1) \left(1 + \frac{1}{p}\right) & 0 \end{array} \right] \quad (10)$$

Solution

$$\xrightarrow{R_2 \rightarrow \frac{1}{(p^2+1)\frac{p+1}{p}} R_2} \left[\begin{array}{cc|c} 1 & -\frac{1}{p(p^2+1)} & 0 \\ 0 & 1 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow R_1 + \frac{1}{p(p^2+1)} R_2} \left[\begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \end{array} \right] \quad (11)$$

For a non-trivial solution $\mathbf{m} \neq \mathbf{0}$, the system must be **rank deficient**, meaning the second row before normalization must be zero:

$$(p^2 + 1) \frac{p + 1}{p} = 0. \quad (12)$$

Solution

Solve for p :

$$p^2 + 1 = 0 \implies p = \pm i \quad (\text{non-real, discard}), \quad (13)$$

or

$$p + 1 = 0 \implies p = -1. \quad (14)$$

Check special case $p = 0$:

If $p = 0$, the matrix becomes

$$\begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix} \quad (15)$$

which has full rank, so only the trivial solution exists.

The only real value of p such that both lines are perpendicular to a common line is

$$p = -1$$

(16)

Code (C)

Part 1

```
#include <stdio.h>
#include <math.h>

int solve_real_p(double *p_real) {
    int count = 0;
    // Factor 1: (p+1) = 0
    p_real[count++] = -1;
    // Factor 2: (p^2 + 1)^2 = 0 -> p^2 + 1 = 0
    // Check for real roots
    double discriminant = -1; // p^2 = -1
    if (discriminant >= 0) {
        p_real[count++] = sqrt(discriminant);
        p_real[count++] = -sqrt(discriminant);
    }
    return count; // number of real roots
}
```


Part 1

```
import ctypes
import os
import matplotlib.pyplot as plt
import numpy as np

# Step 1: Load the C library
if os.name == "nt": # Windows
    lib = ctypes.CDLL("./code11.dll")
else:
    lib = ctypes.CDLL("./code11.so")

from ctypes import import c_double

lib.solve_real_p.argtypes = [ctypes.POINTER(c_double)]
lib.solve_real_p.restype = ctypes.c_int
```

Part 2

```
# Array to store real roots
p_real_array = (c_double * 2)()
num_roots = lib.solve_real_p(p_real_array)

# Use the first real root (p = -1)
p = p_real_array[0]
print(f"Real solution from C function: p = {p}")

# Step 2: Define the lines
q = 1 # arbitrary value
x = np.linspace(-5, 5, 400)

# Line equations
y1 = p*(p**2 + 1)*x + q
y2 = -((p**2 + 1)**2 * x + 2*q)/(p**2 + 1)
```

Python Code: Plotting

Part 3

```
# Step 3: Common perpendicular
# Slope of perpendicular line:  $m_{\text{perp}} = 1/2$ 
m_perp = 1 / 2

# Choose a point halfway between the two lines at x=0
y_mid = (y1[200] + y2[200]) / 2
x_perp = np.linspace(-5, 5, 400)
y_perp = m_perp * x_perp + y_mid # line passing through midpoint

# Step 4: Plot everything
plt.figure(figsize=(7, 5))
plt.plot(x, y1, label='L1:  $p(p^2+1)x - y + q = 0$ ', color='blue')
plt.plot(x, y2, label='L2:  $(p^2+1)^2 x + (p^2+1)y + 2q = 0$ ',
         color='red')
plt.plot(x_perp, y_perp, '--', label='Common perpendicular',
         color='green')
```

Part 4

```
plt.xlabel('x')
plt.ylabel('y')
plt.title(f'Lines and Common Perpendicular for p = {p}')
plt.grid(True)
plt.legend()
plt.axhline(0, color='black', linewidth=0.8)
plt.axvline(0, color='black', linewidth=0.8)

# Step 5: Save figure
plt.savefig("fig11.png", dpi=300)
plt.show()
```

Graphical Representation

